# Fernando Daniel García

## *Personal details*

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## *Current position*

Principal Professional (Research Support Staff)
Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET, *National Scientific and Technical Research Council*)
Centro de Investigaciones del Mar y la Atmósfera (CIMA, *Center for Ocean and Atmospheric Research*)
Buenos Aires - Argentina

CTO and Co-founder
Agilmind SRL
La Plata - Argentina

## *Professional summary*

Computer scientist with experience as a member of research teams in a range of scientific fields. He has developed software and designed innovative technological solutions in mathematics, ecology and, currently, the climate sciences, contributing to the transfer of knowledge to society. He has worked in artificial intelligence since 1984: symbolic artificial intelligence and expert systems (Prolog, LISP and his own inference engine), evolutionary computation (genetic algorithms) and, currently, language models and the integration of artificial intelligence into production systems. In parallel, he was co-founder and CTO of software development companies (XG Tecnología, García & Strauss and FT WAVE), and is co-founder and CTO of Agilmind SRL.

## *Technical skills*

**Programming languages:** Python, JavaScript/TypeScript, C/C++, C#, Java; Prolog and LISP (expert systems).
**Web development and visualization:** Node.js, React, Django, REST APIs, GraphQL; web maps (OpenLayers, MapLibre) and charts (Recharts).
**Data and scientific computing:** NumPy, Matplotlib; processing of satellite data (GOES-16) and forecast data (GFS); automated data collection (Playwright).
**Artificial intelligence:** symbolic, since 1984: expert and rule-based systems in Prolog and LISP, his own inference engine based on Robinson's resolution rule (in C, 1989) and the Augur language for expert systems; evolutionary computation: genetic algorithms (course with David E. Goldberg, Illinois, 2000; GECCO 2000 paper cited in 16 patents; applied in the 2009 game for Axe); currently, AI systems architecture design and orchestration of multiple models in production (LLMs, RAG, neural networks); Hugging Face (Transformers, sentence-transformers, PEFT/LoRA), PyTorch, TensorFlow, pgvector, OpenCV, Tesseract.
**Databases and infrastructure:** PostgreSQL (ORMs: Prisma, Django), Docker, GitHub.

**Domain-specific languages (DSLs):** design and implementation of several DSLs for different kinds of applications; currently Haiku, for full-stack applications, under continuous development.

## *Languages*

Spanish (native), English (intermediate), Portuguese (intermediate).

## *Education*

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| 1983 | Programador Superior (computer programming degree) | Facultad de Ciencias Físico-Matemáticas y Naturales (*School of Physical, Mathematical and Natural Sciences*)<br>Universidad Nacional de San Luis (UNSL, *National University of San Luis*)<br>San Luis - Argentina |

## *Scientific and technological contributions*

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| 2022-2024 | Development of a web application that integrates multiple sources of hydroclimatic information to strengthen people-centered early warning systems, as part of the IMPACT.AR program | CIMA/CONICET-UBA<br>Ministerio de Ciencia y Tecnología de la Nación (MINCYT, *National Ministry of Science and Technology*) |
| 2022 | Development of a web application for recording severe weather events in Argentina, Uruguay, Paraguay, Chile and Brazil. PREVENIR project | Japan International Cooperation Agency (JICA)<br><br>Servicio Meteorológico Nacional (SMN, *National Meteorological Service*) |
| 2020 | Development of the CIMA-GOES library for bulk download and processing of GOES-16 satellite files | CIMA/CONICET-UBA |
| 2020 | Web application for monitoring and forecasting Covid-19 in Argentina | CIMA/CONICET-UBA |
| 2019-2024 | Web application for identifying overshooting tops in satellite images, to train detection methods based on artificial intelligence. | CIMA/CONICET-UBA |
| 2018-2019 | Software development for planning the field experiment:<br>• System that automates the download, processing and publication of GFS data<br>• Scrapers/robots that download data from weather stations in real time | Remote sensing of Electrification, Lightning, And Mesoscale/microscale Processes with Adaptive Ground Observations (RELAMPAGO)<br><br>Clouds, Aerosols, and Complex Terrain Interactions (CACTI) |

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| 2016-2023 | Development of a system of scrapers/robots for collecting data from weather stations, with normalization of the collected data. | CIMA/CONICET-UBA |
| 2015-2026 | Software and visualization development for the Anticipando la Crecida project (community-based flood early warning systems), from its initial microsite to the data tools used in the co-design | CIMA/CONICET-UBA |
| 2014-2019 | Development of CIMAPS, a web platform for visualizing georeferenced information | Centro de Investigaciones del Mar y la Atmósfera (CIMA)<br>CONICET<br>Universidad de Buenos Aires (UBA, *University of Buenos Aires*) |
| 2004 | Development of graphing software for the characterization of sustainability syndromes in development: the case of Argentina. | Sustainable Development and Human Settlements Division<br>Economic Commission for Latin America and the Caribbean (ECLAC)<br>United Nations |
| 2001-2011 | Software development for research in ecology, wildlife management and epidemiology:<br>• Population dynamics and simulation models for the sustainable management of renewable natural resources<br>• Spatial and statistical analysis of field sampling (capybaras, the turquoise-fronted amazon parrot, wild kissing bugs, the southern green stink bug and its parasites)<br>• Geographic distribution and species richness of the triatomine vectors of Chagas disease: climate analyses, neural networks and meta-analysis of life cycles<br>• Computational methods for the sensitivity analysis of dynamic models<br>• Databases of ecological literature and of environmental impact assessment; the Ecopaedia ecological glossary<br>• Robots for downloading scientific literature, georeferenced graphing and digital image processing | Centro de Estudios Parasitológicos y de Vectores (CEPAVE, *Center for Parasitological and Vector Studies*)<br>CONICET<br>Universidad Nacional de La Plata (UNLP, *National University of La Plata*) |
| 2000 | Development of a website to disseminate Property Rights over Land and Ownership of Natural Resources in the Context of Environmental Sustainability in a Market Economy | Lincoln Institute of Land Policy<br>Universidad de Belgrano (UB)<br>Federación Argentina de Comercialización e Industrialización de la Fauna (FACIF, *Argentine Federation for Wildlife Trade and Industry*) |

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| 1995-1999 | Development of an interactive web application implementing population dynamics models | Environmental Software and Services GmbH<br>Gumpoldskirchen - Austria |
| 1995 | Development of a database of species associated with weed control | Weed Biological Control Laboratory<br>United States Department of Agriculture |
| 1994-1995 | Development of the database for the Argentine Patagonian Coastal Zone Management Plan | Global Environment Facility (GEF)<br>World Bank<br>United Nations Development Programme (UNDP) |
| 1989-1993 | Software development:<br>• Inference engine for building expert systems.<br>• LR/LALR parser generator.<br>• Hypertext engine for the Directory of Ecologists and Environmental Scientists of Latin America, the Caribbean and the Iberian Peninsula | Departamento de Biología (*Department of Biology*)<br>Facultad de Ciencias Exactas y Naturales (*School of Exact and Natural Sciences*)<br>Universidad de Buenos Aires (FCEN-UBA) |
| 1988-1989 | Research project: expert system for natural resource management | Dirección de Ecología de la Provincia de Neuquén (*Directorate of Ecology of the Province of Neuquén*) |
| 1987 | Development of a prototype system for monitoring, abnormal event annunciation and control rod interlock logic for the RA3 nuclear research reactor | Comisión Nacional de Energía Atómica (CNEA, *National Atomic Energy Commission*) |
| 1984-1996 | Software development in the following areas:<br>• Network reliability<br>• Graph theory<br>• Symbolic differentiation | Instituto Argentino de Matemática (IAM, *Argentine Institute of Mathematics*)<br>Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET) |

## *Industry experience*

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| Since 2006 | **CTO and Co-founder.** Custom software design and development with in-house tools, with an emphasis on artificial intelligence. | Agilmind SRL<br>La Plata - Argentina |

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| | • **Artificial intelligence:** *Axe Music Star* (2009), an interactive game for Axe: design of the AI system architecture, combining adaptive music, computer vision, genetic algorithms and real-time non-linear video; AgilPost (intelligent accounting allocation of invoices and receipts) and AgilBank (ingestion and reconciliation of bank statements, launching soon): AI architecture design and orchestration of several models, the basis of Pronto IAC and Pronto IEB, marketed by BDL Consultores; AgilFace (facial recognition), for companies operating in Vaca Muerta.<br>• **Other products:** AgilSign (electronic signature); AgilCred (personal loan management for Frávega's branches); AgilLic (tender management) and AgilDoc (document tracking with expiry alerts), marketed by BDL Consultores; Jupiter, a platform for the music industry and copyright, successor to Plutón (automation for music publishers, 2024; client: Sony Music Publishing), in which he took part as CTO.<br>• **Standards:** JDX (Jupiter Data eXchange), a specification of a data exchange protocol for registering musical works with collective management organizations, of which he is the author (Zenodo, 2026, https://doi.org/10.5281/zenodo.23137511).<br>• **Clients:** FOX Play, Primestream Corp, Unilever, Tuscany Now & More, Prodigy Network, Frávega, Autotrol, Grupo Clarín, Sony Music Publishing, FPA, BDL Consultores, SAM Sistemas.<br>• **Predecessor:** FT WAVE (2000–2002), which Agilmind continues: an online system for distributing new music releases to radio stations across Argentina, for Sony Music, EMI Music, Warner Music, BMG, Universal Music and Leader Music. | |
| 2001-2006 | **Consultant.** Development of a rapid application development (RAD) framework for Delphi and design of multi-tier architectures. | Borland Argentina<br>Buenos Aires |

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| 2000-2002 | **CTO and Co-founder.** Design of an online system for distributing new music releases to every radio station in the country over dial-up connections, with its own digital music player, aimed at curbing piracy. Clients: Sony Music, Warner, BMG, EMI, Universal Music, Leader Music. | FT WAVE S.A. |
| 1995-2004 | **CTO and Co-founder.** Software design and development focused on user interfaces. Clients: Siemens, TPS, FPA Software, Car One, Sirsa, Supermercados Disco, Norte, Tigre; AFJP Previnter, AFJP Allianz, Grupo Siembra, Cenit Bansud; Benito Roggio, Rennell, Twice Technology, Paseo La Plaza, Teatro Avenida, Teatro Colón de Mar del Plata. | García & Strauss S.H. |
| 1993-1995 | **CTO and Co-founder.** Pioneering design and development of multimedia software in Argentina. Clients: Ford, Volkswagen, Toyota, Sevel, Autolatina, Bagley, Sancor, Nabisco-Terrabussi, Marlboro, Johnson, Kimberly Clark, Telecom. | XG Tecnología S.H. |
| 1983-1984 | **Programmer.** Programming in the Mantis language with the Total database for an integrated banking system. | Technologica S.A. |

### *Research projects*

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| 2022-2026 | The project for numerical weather prediction and warning communication system for densely populated and vulnerable cities | Japan International Cooperation Agency (JICA) |
| 2022-2024 | Very short-range weather forecasting using radar data assimilation and machine learning | Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET) |
| 2022-2024 | People-centered early warning system in the Municipality of La Matanza, for the IMPACT.AR program. | Ministerio de Ciencia, Tecnología e Innovación (MINCYT, *Ministry of Science, Technology and Innovation*)<br>CIMA/CONICET-UBA<br>Municipio de La Matanza |
| 2018-2022 | RELAMPAGO: an opportunity to improve the understanding of the physical and dynamical processes that initiate and sustain high-impact weather events<br>International field campaign (Argentina, the United States and Brazil; June 2018 – April 2019). Contributions: real-time collection of weather station data (Argentina Mesonet Data dataset, NCAR) and processing of GFS forecasts for operations planning. | Departamento de Ciencias de la Atmósfera y los Océanos (DCAO, *Department of Atmospheric and Ocean Sciences*)<br>Facultad de Ciencias Exactas y Naturales<br>Universidad de Buenos Aires (FCEN-UBA) |

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| 2015-2026 | Anticipando la Crecida: co-design and implementation of community-based flood early warning systems in La Matanza, Lomas de Zamora, Avellaneda and Lanús (Buenos Aires Metropolitan Area). Contributions: data curation, methodology, software development and visualization. | Centro de Investigaciones del Mar y la Atmósfera (CIMA)<br>CONICET<br>Universidad de Buenos Aires (UBA) |
| 1995-1999 | GAIA: A Multi-Media Tool for Natural Resources Management and Environmental Education | Environmental Software and Services GmbH<br>Gumpoldskirchen - Austria |
| 1994-1995 | PMZCP: Argentine Patagonian Coastal Zone Management Plan | Global Environment Facility (GEF)<br>Argentina |

## *Publications*

Simone, I., Salio, P., Ruiz, J., Cancelada, M., Vidal, L. and **García, F.** (in preparation). A short climatology of overshooting cloud top detection developed from a simplified method in southern South America.

Robledo, F., Moreira, D., Prudente, C., … **García, F.**, … Pérez Vega, A. E. [36 authors] (2026). Co-design and implementation of community-based flood early warning systems in eastern Argentina. *Environmental Research: Climate*, *5*(3), 035041. https://doi.org/10.1088/2752-5295/ae7fac

Salio, P., Bechis, H., Ribeiro, B. Z., de Lima Nascimento, E., Galligani, V., **García, F.**, … Villagrán Asiares, C. I. [39 authors] (2024). Toward a South American High-Impact Weather Reports Database. *Bulletin of the American Meteorological Society, 105*(7), E1204–E1217. https://doi.org/10.1175/BAMS-D-23-0063.1

**García, F.**, Ruiz, J., Salio, P., Bechis, H. and Nesbitt, S. (2019). *Argentina Mesonet Data* (Version 1.1) [Data set]. UCAR/NCAR – Earth Observing Laboratory. https://doi.org/10.26023/JEXB-W6B6-E310. Also in CONICET's research data repository (2024): http://hdl.handle.net/11336/243011

**García, F.** (2000). Computer Screen Design Aided by a Genetic Algorithm. In D. Whitley (Ed.), *Late Breaking Papers at the 2000 Genetic and Evolutionary Computation Conference* (pp. 98–101).

*Cited as prior art in 16 US patents from a single line of technology (2006–2023), assigned to Affinnova, Inc. and Nielsen; among them, US 7,016,882 and US 9,785,995.*

## *Presentations at conferences and meetings*

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| 2022 | "Towards a South American severe storm database". Co-author, with H. Bechis, B. Ribeiro, P. Salio and others. | 30th Conference on Severe Local Storms<br>American Meteorological Society (AMS)<br>Santa Fe, New Mexico - USA |
| 2022 | "Hacia la creación de una base de datos de tormentas severas en Sudamérica" (*towards a severe storm database for South America*). Co-author, with H. Bechis, B. Zanetti Ribeiro, P. Salio and others. Abstract: http://hdl.handle.net/20.500.12160/2474 | XIV Congreso Argentino de Meteorología (CONGREMET XIV, *14th Argentine Congress of Meteorology*)<br>Centro Argentino de Meteorólogos<br>Buenos Aires - Argentina |

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| 1996 | "Argentina case study" | GAIA Project Startup - Meeting<br>Environmental Software and Services GmbH<br>Gumpoldskirchen - Austria |
| 1991 | "Sistema Experto para el Manejo del Zorro Colorado de Patagonia" (*an expert system for the management of the Patagonian red fox*) | 3er Encuentro Regional de Ingeniería de Sistemas & Workshop CLEI Sistemas Expertos y Multimedios (*3rd Regional Systems Engineering Meeting and CLEI Workshop on Expert Systems and Multimedia*)<br>Concepción - Chile |

## *Courses and seminars taught*

| | | |
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| 2001 | • Application development with Delphi<br>• Multi-tier application design with WebSnap and DataSnap | Borland Argentina<br>Buenos Aires |
| 1991 | Introduction to Compiler Theory | Estudio Weber, Ferro & Asociados<br>Buenos Aires |
| 1985 | Seminar on "Descriptive Logic Languages and Functional Programming" | Instituto Argentino de Matemática |
| 1985 | Co-taught the theoretical and practical course "Databases for the Management and Conservation of Wildlife and Protected Natural Areas" | Secretaría de Agricultura, Ganadería y Pesca de la Nación (*National Secretariat of Agriculture, Livestock and Fisheries*)<br>Ministerio de Economía (*Ministry of Economy*)<br>Buenos Aires |
| 1984 | Co-taught the graduate seminar on "Information Systems and Their Application to Urban and Regional Planning" | Facultad de Arquitectura y Urbanismo (*School of Architecture and Urban Planning*)<br>Universidad de Buenos Aires |

## *Courses and further training*

| | | |
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| 2000 | Genetic Algorithms in Search, Optimization, and Machine Learning | University of Illinois at Urbana-Champaign |
| 1987 | • Design of Language-Centered Environments<br>• Knowledge Engineering and Expert Systems<br>• Computerized Industrial Automation | Escuela Argentino Brasileña de Informática (EBAI, *Argentine-Brazilian School of Computer Science*)<br>Tandil - Argentina |
| 1986 | • Applied Artificial Intelligence<br>• Theory of Computation<br>• Program Verification and Derivation | Escuela Argentino Brasileña de Informática (EBAI)<br>Campinas - Brazil |

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| 1984 | • Local Area Network Technology<br>• Microprocessor Architecture<br>• LSI: Structured Design of Integrated Circuits | Centro Latinoamericano de Matemática e Informática (CLAMI, *Latin American Center for Mathematics and Computer Science*)<br>Buenos Aires - Argentina |
| 1983 | Introduction to Reliability | Instituto de Matemática Pura e Aplicada (IMPA, *Institute for Pure and Applied Mathematics*)<br>Rio de Janeiro - Brazil |

### *Fellowships*

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| 1983 | Research initiation fellowship | Instituto de Matemática Pura e Aplicada (IMPA)<br>Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq, *National Council for Scientific and Technological Development*)<br>Rio de Janeiro - Brazil |

### *Memberships*

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| 1996-2016 | Member | Association for Computing Machinery (ACM)<br>New York - USA |
| 2000-2001 | Member. The society, which ran the GECCO conference, became ACM SIGEVO in 2005. | International Society for Genetic and Evolutionary Computation (ISGEC)<br>Stanford, California - USA |

### *Music*

He has sung in folk groups since the age of 12 (among them Inti Huasi, which he founded), was a member of the Grupo de Experimentación Vocal (GEV), wrote arrangements for the Fundación San Telmo choir and has composed music for theatre, such as *Vanya* (2025), directed by José María López (Kusmis Teatro); earlier, for *El tuerto es rey* by Carlos Fuentes (1987) and *Kaspar* by Peter Handke (1991), both directed by Juan Polito. Solo album: *Ronda* (2015). In 2005 he developed the software for *Digidramamancia*, which generated the script and the score in real time from the audience's fingerprints, and was responsible for its metacomposition and musical direction; the play was also directed by José María López. Registered as author, performer and phonogram producer with SADAIC, AADI and CAPIF. Music at https://fidogarcia.bandcamp.com.